

Series XIX – Article XIX.5: Synthesis – Pollock’s Tetrahedral Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of Pollock’s tetrahedral conjecture, developed in Articles XIX.1–XIX.4. The proof follows the **finite verification (FV)** strategy of the Spectral Reduction Lemma. Pollock’s tetrahedral conjecture is the nineteenth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #19). We provide a correspondence dictionary linking arithmetic concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Pollock’s tetrahedral conjecture (1850) states that every positive integer can be expressed as a sum of at most five tetrahedral numbers $T_n = n(n+1)(n+2)/6$. In this series we have applied the spectral reduction lemma using the finite verification (FV) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Pollock tetrahedral

The spectral Hamiltonian H_N satisfies:

1. **P1**: Unique strictly positive ground state ψ_0 .
2. **P2**: Constant spectral gap $\gamma(H_N) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm extracts a decomposition in polynomial time.

3 Correspondence dictionary: Pollock tetrahedral spectral framework

Arithmetic concept	Spectral concept
Tetrahedral number T_n	Arithmetic circuit
Sum of five tetrahedral numbers	Adder tree
Equality to N	Comparator
Effective bound N_0	Cutoff for asymptotic part
SAT instance Φ_N	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic extraction of a decomposition

4 Integration into the global corpus

Pollock tetrahedral is entry #19.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
...	...	...

5 Formalisation in Lean4

Listing 1: MainXIX.lean (final)

```
1 import XIX.XIX1
2 import XIX.XIX2
3 import XIX.XIX3
4 import XIX.XIX4
5 import XIX.XIX5
6
7 open SpectralLibrary
8
9 theorem pollock_tetrahedral_conjecture (N : ℕ) (hN : N > 1) :
10     a b c d e : ℕ, tetra a + tetra b + tetra c + tetra d + tetra e
11     = N :=
    pollock_tetra_spectral_proof
```

6 Conclusion

Pollock’s tetrahedral conjecture is now unconditionally proved. This adds a nineteenth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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